SLIDE-QUIZ EXERCISE · ATMQS01

Atmospheric Pressure

Understanding the Pressure of the Air Around Us

TEACHER GUIDE

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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EXERCISE OVERVIEW

Atmospheric Pressure

This exercise explores the weight of the air we live beneath. Four content pages move from the idea of pressure itself to the way it was first measured and the many units used to report it today. The first page defines pressure and atmospheric pressure and explains why the air presses on us at all; the second follows Torricelli's mercury barometer to the modern digital sensor; and the last two sort out the units, from the Pascal and the millibar of the weather map to the psi of a car tire, the mm Hg of a blood-pressure reading, and the in Hg of an aircraft altimeter. Every content page has a narrated instruction panel you can play, two pages have interactive figures, and the graded questions number six in all.

The Exercise, Page by Page

Page	What you'll do
Introduction	A short video explains what to expect on the pages that follow.
1. What Is Pressure?	Pressure as force divided by area, and atmospheric pressure as the weight of the air above you. Two questions.
2. Measuring Atmospheric Pressure	Torricelli's mercury barometer, the barometer simulation, and modern barometers. Two questions.
3. Pressure Units — Part 1	Pascal, Atmosphere, bar, millibar, and hectopascal; Blaise Pascal. One question.
4. Pressure Units — Part 2	psi, mm Hg (Torr), and in Hg; blood pressure and the aircraft altimeter. One question.
Conclusion	A closing video, your score, and the Submit Quiz button.

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Learning Objectives

After this exercise you should be able to:

- Define pressure as force divided by area, and state its metric (SI) unit, the Pascal.
- Define atmospheric pressure, and explain why it decreases with increasing altitude.
- Describe Torricelli's mercury barometer and explain what holds the mercury column up.
- State the general relationship between atmospheric pressure and the weather.
- Name the modern instruments that have replaced the mercury barometer.
- Give the value of one Atmosphere in Pa, bar, mbar, hPa, psi, mm Hg, and in Hg, and convert between them.
- Identify where each unit is used: weather maps, tires, medicine, and aviation.

QUICK REFERENCE

Key Terms

Term	Definition
Pressure	The force applied perpendicularly to a surface divided by the area of that surface. Pressure is a scalar quantity.
Atmospheric pressure	The weight of the column of air directly above a given altitude, divided by the cross-sectional area of that column.
Pascal (Pa)	The metric (SI) unit of pressure: one Newton of force spread over one square meter. A very small amount of pressure.
Barometer	An instrument that measures atmospheric pressure.
Mercury barometer	Torricelli's 1643 instrument: a glass tube of mercury inverted into a dish of mercury. The air pressing on the dish holds up a column about 760 mm tall at sea level.
Torricellian vacuum	The empty space above the mercury at the top of the tube — the first sustained man-made vacuum.
Aneroid barometer	A mechanical barometer with a sealed, flexible metal capsule that expands and contracts as the pressure changes (Lucien Vidie, 1844).
Digital (MEMS) barometer	An electronic barometer whose tiny pressure-sensing chip changes its electrical properties as the pressure changes; found in phones, weather stations, and aircraft.
Barograph	A recording barometer that traces pressure continuously on paper wrapped around a rotating drum.
Atmosphere (Atm)	A non-SI unit equal to the average air pressure at sea level: 1 Atm = 101,325 Pa.
Bar / millibar (mbar, mb)	1 bar = 100,000 Pa; 1 mbar = 1/1000 bar = 100 Pa. The millibar was the standard weather unit for most of the 20th century.
Hectopascal (hPa)	100 Pa. Numerically identical to the millibar; the unit preferred by the World Meteorological Organization.

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Term	Definition
Pascal's Principle	An external pressure applied to a liquid is extended, undiminished, to every point in the liquid and to the walls of its container.
psi	Pounds per square inch: one pound of force on one square inch. Used for tire, scuba-tank, and industrial pressures in the US.
mm Hg / Torr	The pressure that supports a 1 mm column of mercury. The Torr (for Torricelli) is 1/760 of an Atmosphere. Used for blood pressure.
in Hg	Inches of mercury. Used for US weather reports and for setting aircraft altimeters.
Altimeter	An aircraft instrument that reads altitude from the outside (static) air pressure, which falls as the aircraft climbs.
Kollsman window	The small window on an altimeter where the pilot dials in the local barometric pressure (in in Hg in the US).

QUICK REFERENCE

The Units at a Glance

Unit	Symbol	1 Atm equals	Where it's used
Pascal	Pa	101,325 Pa	The metric (SI) unit; science
Atmosphere	Atm	1 Atm (the reference)	Comparing any pressure with sea level
Bar	bar	1.01325 bar	Oceanography, meteorology, industry
Millibar	mbar, mb	1,013.25 mbar	Weather maps, aviation weather
Hectopascal	hPa	1,013.25 hPa	Modern weather reports (WMO standard)
Pounds per square inch	psi	14.696 psi	Tires, scuba tanks, hydraulics (US)
Millimeters of mercury	mm Hg, Torr	760 mm Hg = 760 Torr	Blood pressure, vacuum work
Inches of mercury	in Hg	29.921 in Hg	US weather reports, altimeter settings

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QUICK REFERENCE

Numbers to Know

Value	What it refers to
101,325 Pa	One Atmosphere: average air pressure at sea level
14.7 psi	The same pressure in pounds per square inch (14.696 precisely)
760 mm (29.92 in)	Height of Torricelli's mercury column at sea level
1 Pa	One Newton of force on one square meter: *1 Pa = 1 N / 1 m²*
100,000 Pa	One bar
100 Pa	One millibar, which equals one hectopascal
1.3%	How much greater one Atmosphere is than one bar
1643	Torricelli invents the mercury barometer
1648	Puy de Dôme experiment shows pressure falls with altitude
1844	Lucien Vidie's aneroid barometer
870–900 mbar	Pressure near the eye of a very powerful hurricane
30–35 psi	Typical car tire pressure
1,020 mb and up	"Fair" on the barometer simulation; below 1,000 mb it reads Rain, and below 980 mb storms

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EXERCISE CONTENT

Page-by-Page Summary

Part 1 — What Is Pressure?

Pressure is the force applied perpendicularly to a surface divided by the area of that surface, $P = F / A$. It is a scalar: it has a size but no direction. Pressure depends on both the force and the area it acts on, so a small force on a tiny area makes a very large pressure — a needle pierces skin where a broad palm cannot. The metric (SI) unit is the Pascal, $1\ Pa = 1\ N / 1\ m^2$, a very small amount of pressure.

Atmospheric pressure is the weight of the column of air directly above a given altitude, divided by the area of that column. We live at the bottom of an ocean of air; the higher you go, the less air is above you and the lower the pressure. At sea level it is 101,325 Pa, or 14.7 pounds on every square inch. We don't feel it because the air inside our bodies, in the lungs, pushes outward just as hard.

Part 2 — Measuring Atmospheric Pressure

In 1643 Evangelista Torricelli (1608–1647), a student and successor of Galileo, filled a glass tube with mercury and turned it upside down into a dish of mercury. The column settled at about 760 mm, or 29.92 inches. A vacuum does not "pull" it up: the weight of the atmosphere pressing on the open dish holds the column up, so its height is a direct measure of the air pressure. The empty space at the top, the Torricellian vacuum, was the first sustained man-made vacuum. "We live submerged at the bottom of an ocean of air," he wrote.

The barometer simulation shows the column rising and falling with the pressure, and the forecast moving from Fair at high pressure to Rain and storms at low pressure: the higher the pressure, the fairer the weather.

Because mercury is extremely toxic, it has been replaced by the aneroid barometer (a sealed, flexible metal capsule, 1844), the electronic or digital barometer (a tiny MEMS pressure chip, found in smartphones, aircraft, and weather stations), and the barograph, which records pressure on a rotating paper drum.

Part 3 — Pressure Units, Part 1

Every unit describes the same quantity, measured against one reference: 1 Atm = 101,325 Pa. The Pascal, named for Blaise Pascal, is so small that larger units are often handier. The bar (1909) is exactly 100,000 Pa, so 1 Atm = 1.01325 bar, only 1.3% more. The millibar is a thousandth of a bar, or 100 Pa, so 1 Atm = 1,013.25 mbar. The hectopascal is also 100 Pa — numerically identical to the millibar — and is the World Meteorological Organization's preferred unit.

The units are linear, so converting is just multiplying: $0.5\ Atm = 50{,}662.5\ Pa = 506.625\ mbar$.

Blaise Pascal (1623–1662) showed with the 1648 Puy de Dôme experiment that air pressure falls with altitude, built an early mechanical calculator, and gave his name to Pascal's

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Principle: a pressure applied to a liquid is passed, undiminished, to every point in the liquid and to the walls of its container.

Part 4 — Pressure Units, Part 2

Three units are especially common in the United States. Pounds per square inch, with 1 Atm = 14.696 psi, is used for tires (a typical car runs 30–35 psi), scuba tanks, and hydraulics. Millimeters of mercury come straight from Torricelli's barometer: 1 Atm = 760 mm Hg = 760 Torr, and blood pressure is always reported in mm Hg. Inches of mercury, with 1 Atm = 29.921 in Hg, are used in US weather reports and to set aircraft altimeters. Again: *0.5 Atm = 7.35 psi = 380 mm Hg = 14.96 in Hg*.

The altimeter diagram shows how a plane knows its height. Its static port senses only the outside air pressure; as the plane climbs, that pressure falls and the altimeter needle turns. The pilot dials the local pressure into the Kollsman window so the altimeter reads the correct altitude.

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STUDY QUESTIONS

Questions with Model Answers

Question numbering matches the student study guide exactly. Model answers are written at the level expected of a strong student response, not as the only acceptable wording.

Red numbers mark core questions — the ones to be sure you can answer.

Part 1 — What Is Pressure?

1. Define pressure in words, and write the equation for it.

ANSWER Pressure is the force applied perpendicularly to a surface divided by the area of that surface: $P = F / A$. It is a scalar quantity (it has size but no direction).

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2. What is the metric (SI) unit of pressure, and how is it defined? Is it a large or a small amount of pressure?

ANSWER The Pascal (Pa), named for Blaise Pascal. $1\ Pa = 1\ N / 1\ m^2$ — one Newton of force spread over one square meter. It is a very small pressure; sea-level air pressure is over 100,000 Pa.

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3. Using $P = F / A$, explain why a sharp needle can pierce skin while a broad palm pushing with the same force cannot.

ANSWER The force is the same, but the needle point has a tiny area. Dividing the same force by a much smaller area gives a much larger pressure — large enough to break the skin. The palm spreads the force over a large area, so the pressure is small.

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4. Define atmospheric pressure.

ANSWER The weight of the column of air directly above a particular altitude, divided by the cross-sectional area of that column.

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5. How does atmospheric pressure change as you go higher, and why?

ANSWER It decreases with increasing altitude. The higher you are, the less air there is above you, so the air column above you weighs less and presses down less.

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6. At sea level the atmosphere pushes on every square inch of your body with about 14.7 pounds of force. Why don't you feel it?

ANSWER The air inside your body, especially in your lungs, pushes outward with an equal pressure, so the inward and outward pushes balance.

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Part 2 — Measuring Atmospheric Pressure

1. Who invented the mercury barometer, and in what year? What was his connection to Galileo?

ANSWER Evangelista Torricelli, an Italian physicist and mathematician, in 1643. He was a student and successor of Galileo.

2. Describe Torricelli's experiment. At what height did the mercury column settle at sea level, in millimeters and in inches?

ANSWER He filled a glass tube, closed at one end, with mercury and turned it upside down into an open dish of mercury. The column dropped and settled at about 760 mm (76 cm), or 29.92 inches.

3. What holds the mercury column up in the tube? Is the vacuum at the top "pulling" it up?

ANSWER No. The weight of the atmosphere pressing down on the open dish of mercury pushes the mercury up the tube and supports the column. The column's height is a direct measure of the air pressure.

4. What is the empty space above the mercury called, and why was it historically important?

ANSWER The Torricellian vacuum. It was the first sustained man-made vacuum ever created.

5. In the barometer simulation, what happens to the mercury column and to the forecast when you raise and lower the pressure? State the general rule linking pressure and weather.

ANSWER Raising the pressure makes the column taller and the forecast moves toward Fair; lowering it shortens the column and the forecast moves to Uncertain, Rain, and finally severe storms. The higher the pressure, the fairer (sunnier) the weather; low pressure brings rain and storms.

6. Why are mercury barometers no longer used? Name the three modern instruments on this page and say briefly how each works.

ANSWER Mercury is extremely toxic. Aneroid barometer: a sealed, flexible metal capsule expands and contracts with pressure, and linkages move a pointer. Digital (electronic) barometer: a tiny MEMS pressure chip changes its electrical signal with pressure, which is calibrated and displayed. Barograph: a recording barometer that traces pressure on a rotating paper drum.

Part 3 — Pressure Units, Part 1

1. What is one Atmosphere (Atm), and what is its value in Pascals?

ANSWER The average air pressure at sea level, used as the standard reference. 1 Atm = 101,325 Pa.

2. Give the value of 1 Atm in bar, in millibars, and in hectopascals.

ANSWER 1.01325 bar; 1,013.25 mbar; 1,013.25 hPa.

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3. How many Pascals are in one millibar and in one hectopascal? Why was switching weather maps from millibars to hectopascals so easy?

ANSWER Both equal 100 Pa. Because 1 hPa = 1 mbar, the numbers did not change — 1,013.25 mbar is exactly 1,013.25 hPa — so only the unit label changed.

4. All pressure units are linear. Convert 0.5 Atm into Pascals and into millibars. Show your work.

ANSWER *0.5 × 101,325 Pa = 50,662.5 Pa*; *0.5 × 1,013.25 mbar = 506.625 mbar*.

5. Who was Blaise Pascal? What did the 1648 Puy de Dôme experiment show, and what does Pascal's Principle say?

ANSWER A French physicist, mathematician, and inventor (1623–1662) who also built an early mechanical calculator; the SI unit of pressure is named for him. The Puy de Dôme experiment showed that atmospheric pressure decreases with altitude. Pascal's Principle: an external pressure applied to a liquid is extended, undiminished, to every point in the liquid and to the walls of its container.

6. Near the eye of a very powerful hurricane the pressure can drop to about 900 mbar. Roughly what fraction of one Atmosphere is that?

ANSWER *900 / 1,013.25 ≈ 0.89* — about 89% of normal sea-level pressure, or about 0.89 Atm.

Part 4 — Pressure Units, Part 2

1. What does psi stand for? What is 1 Atm in psi, and where is psi used?

ANSWER Pounds per square inch. 1 Atm = 14.696 psi (often rounded to 14.7). Used in the US for tire pressure (a typical car: 30–35 psi), scuba tanks, and industrial hydraulics and pneumatics.

2. What is a millimeter of mercury, and how is it connected to Torricelli's barometer? How does it compare to the Torr?

ANSWER The pressure that supports a 1 mm tall column of mercury — it comes straight from the height of Torricelli's column. The Torr, named for Torricelli, is 1/760 of an Atmosphere and is effectively the same as the mm Hg: 1 Atm = 760 mm Hg = 760 Torr.

3. In what unit is blood pressure reported?

ANSWER Millimeters of mercury (mm Hg), for both the systolic and diastolic readings.

4. What is 1 Atm in inches of mercury, and where is in Hg the standard unit?

ANSWER 29.921 in Hg (about 29.92). It is standard for US weather reports and TV forecasts and for setting aircraft altimeters in the US.

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5. How does an aircraft altimeter use air pressure to show altitude? What is the Kollsman window for?

ANSWER The static port senses only the outside air pressure. As the plane climbs, that pressure falls, and the altimeter converts the drop into feet of altitude, turning the needle clockwise. The pilot dials the local barometric pressure into the Kollsman window so the altimeter reads the correct altitude.

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6. Convert 0.5 Atm into psi, mm Hg, and in Hg.

ANSWER *0.5 × 14.696 ≈ 7.35 psi*; *0.5 × 760 = 380 mm Hg*; *0.5 × 29.921 ≈ 14.96 in Hg*.

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ANSWER KEY

The 6 Graded Slide-Quiz Questions

The questions students answer inside the exercise, in page order.

Page	Q	Question	Keyed answer
1. Pressure	1	How is pressure defined?	C) Force divided by area
1. Pressure	2	How does atmospheric pressure change with altitude?	B) It decreases with increasing altitude
2. Measuring	3	Who invented the mercury barometer?	D) Torricelli
2. Measuring	4	Relationship between pressure and weather?	A) The higher the pressure, the fairer the weather
3. Units, Part 1	5	How many mb equal one Atm?	C) 1013.25
4. Units, Part 2	6	Unit used to measure blood pressure?	A) Millimeters of mercury (mm Hg)

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TEACHER NOTES

Corrections and Cautions

Points in the exercise that need a correction, a qualification, or a prepared answer for the question a sharp student will ask.

C1 **“Density” should be “pressure.”**

Page 4’s Real-world check box reads “The sea-level density of air is commonly stated as 101,325 Pa = 14.7 psi.” Those are pressures; sea-level air density is about 1.2 kg/m³. A one-word fix on the page.

C2 **One Atmosphere has no temperature attached.**

The narration for pages 1 and 2 gives 101,325 Pa “at sea level and 20 Celsius,” and the page 4 table heading says “1 standard Atmosphere (NTP = 20°C, sea level).” The Atmosphere is a defined value, exactly 101,325 Pa. NTP is a reference for gas samples (20°C at 1 Atm), and the International Standard Atmosphere uses 15°C at sea level. Students may come away thinking temperature is part of the definition; the study guide gives 1 Atm without one.

C3 **Only the Pascal is SI.**

The introduction lists “Metric units: Pascal, Atmospheres, bar, and millibar,” and the conclusion says “Metric (SI) pressure units: Pa, Atm, bar, millibar (mbar), and hectopascal (hPa).” The Pascal and its prefixed forms (hPa) are SI; the bar and millibar are metric but not SI; the Atm is non-SI, as page 3 itself says.

C4 **PDS means “Particularly Dangerous Situation.”**

Graded question 4’s option D and the barometer simulation’s lowest forecast use “PDS.” In National Weather Service usage it stands for Particularly Dangerous Situation, not “particularly damaging storms.” Option D is still wrong for the right reason — tornadoes and severe storms go with low pressure — so the keyed answer is unaffected.

C5 **Digital barometers are not usually piezoelectric.**

The page 2 narration says a digital barometer works through a piezoelectric crystal; the page says “a piezoelectric crystal or MEMS chip that changes its electrical resistance.” Nearly all barometric MEMS sensors — including the BME280 in Lab Activity 2 — are piezoresistive or capacitive. Piezoelectric sensors respond only to changes in pressure and cannot hold a steady reading. The study guide says “a tiny MEMS pressure chip.”

C6 **What holds the mercury up.**

The page 2 narration explains the column with Pascal’s Principle. The fuller explanation is a balance: the column rises until the weight of mercury per unit area in the tube equals the air pressure on the open dish. Pascal’s Principle explains how the pressure on the dish is passed through the mercury to the base of the column. The study guide uses the balance wording.

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C7 **Who climbed the Puy de Dôme.**

Page 3 says Pascal demonstrated the fall of pressure with altitude in "an experiment carried out on Puy de Dôme in 1648." That is accurate: Pascal designed the experiment, but his brother-in-law Florin Périer carried the barometer up the mountain.

C8 **Torricelli as Galileo's "student."**

Torricelli studied Galileo's work and was his assistant during the last months of Galileo's life (1641–42), then succeeded him as mathematician to the Grand Duke of Tuscany. "Student and successor" is fair shorthand, and graded question 3's distractor B (Galileo) is built on this link.

C9 **Small rounding and wording points.**

The Fig. 3 caption gives 14.69 psi; 14.696 rounds to 14.70. The Fig. 6 caption's "true altitude" is loose: with the local setting dialed in, the altimeter reads height above sea level, which can still be off when the air is much warmer or colder than standard. The conclusion page heading reads "You've Completed the Atmospheric Pressure Module"; the guides call it an exercise.

TEACHER NOTES

Teaching the Exercise

Run the barometer slider together.

Start at 1,013.25 mb (Reset to 1 Atm), then slide to about 1,030 mb (Fair), 1,010 mb (Uncertain), 990 mb (Rain), and 960 mb. Have students watch the column height and the Pa, Atm, and psi readouts change in step — a live demonstration that the units are linear, and a setup for graded question 4.

Pose the altimeter question.

If a pilot flies from high pressure into low pressure without resetting the Kollsman window, does the altimeter read too high or too low? Too high — the plane is lower than the instrument shows. Pilots remember it as "high to low, look out below." It ties pressure, altitude, and safety into one question.

Listen for the sucking vacuum.

The most common misconception is that the vacuum at the top of the tube pulls the mercury up. Turn it around: nothing pulls; the air pushes down on the dish. Two related slips: that air pressure only pushes downward (it acts in all directions, which is why we are not crushed), and that hPa and mbar are different amounts (they are numerically identical).

Ask how tall a water barometer would be.

Mercury is 13.6 times denser than water, so a water barometer needs a column about 13.6 times taller: roughly 10.3 m, or 34 feet. It makes the point that the air's push is large, and it explains why Torricelli chose mercury.

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Tie it back to astronomy.

Surface pressure is one of the first things we want to know about another world. Mars has less than 1% of Earth's surface pressure, too little for liquid water to last on its surface; Venus has about 90 times Earth's, like being nearly a kilometer under the ocean. Students can express both in the units from this exercise.

Connect it to Module 5A and Lab Activity 2.

Module 5A shows how pressure falls through the layers of the atmosphere, and the Weather Lab's BME280 sensor reports pressure in hPa — students can check its reading against the numbers here.

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